

11301 微積分乙期末考評分標準

題號 (1)

$$\lim_{x \rightarrow 0} \frac{x^2}{\ln(\cos 2x)} \quad \left(\frac{0}{0} \right)$$

一開始微錯全扣 (分子微對可得1分)

$$\stackrel{(L')}{=} \lim_{x \rightarrow 0} \frac{2x}{\frac{1}{\cos 2x} \cdot (-\sin 2x) \cdot 2} \quad 2 \text{分}$$

第一次羅必達 2分 (微錯扣一分)

第二次羅必達 2分 (微錯扣一分)

$$\stackrel{(法一)}{=} \lim_{x \rightarrow 0} \frac{x}{-\tan 2x} \quad \left(\frac{0}{0} \right)$$

$$\stackrel{(L')}{=} \lim_{x \rightarrow 0} \frac{1}{-\sec^2 2x \cdot 2} \quad 2 \text{分}$$

$$= -\frac{1}{2}$$

2分

(法二)

$$\lim_{x \rightarrow 0} \frac{x^2}{\frac{-\sin 2x}{\cos 2x} \cdot 2} = \lim_{x \rightarrow 0} \frac{x \cdot \cos 2x}{-\sin 2x} \quad \left(\frac{0}{0} \right)$$

$$\stackrel{(L'H)}{=} \lim_{x \rightarrow 0} \frac{\cos 2x + x(-\sin 2x \cdot 2)}{-2 \cos 2x}$$

$$= -\frac{1}{2}$$

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題號 1. (2)

$$1. (2) \lim_{x \rightarrow 0^+} (e^{2x} + 2x)^{\frac{2}{x}} \quad (1^{\infty})$$

$$\lim_{x \rightarrow 0^+} \ln (e^{2x} + 2x)^{\frac{2}{x}} \quad \text{2分 (可直接寫第=行)}$$

$$= \lim_{x \rightarrow 0^+} \frac{2 \ln (e^{2x} + 2x)}{x} \quad \left(\frac{0}{0} \right)$$

$$\stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \frac{2 \cdot \frac{2e^{2x} + 2}{e^{2x} + 2x}}{1} \quad \text{2分}$$

$$= 8$$

$$\text{Ans: } \underline{e^8} \quad \text{2分 (最後答案 2分)}$$

* 若最後答案只寫 8, 則扣 1 分,

* 過程不對, 答案對不給分.

題號 1. (3)

$$\lim_{x \rightarrow 0^+} \frac{\int_{\sqrt{x}}^0 \sin(t^2) dt}{\sqrt{x^3}}$$

$$\stackrel{L'}{=} \lim_{x \rightarrow 0^+} \frac{-\sin x \cdot \frac{1}{2\sqrt{x}}}{\frac{3}{2}\sqrt{x}} \quad \text{2分}$$

不扣分

$$= \lim_{x \rightarrow 0^+} \frac{-1}{3} \cdot \frac{\sin x}{x} = \underline{\underline{-\frac{1}{3}}} \quad \text{1分} \quad \text{1分}$$

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題號 2, 4, 8.

(1). $\int \frac{\cos^5 x}{\sin^2 x} dx.$

①. $u = \sin x.$
 $du = \cos x dx.$ +2.

* $\sin x \neq \frac{1}{\sin x}$
 有筆誤會多扣 1 分
 扣到 0 分為止.

$= \int \frac{\cos^4 x}{\sin^2 x} \cos x dx = \int \frac{(1-u^2)^2}{u^2} du.$ +1.

$= \int u^2 - 2 + \frac{1}{u^2} du = \frac{1}{3} u^3 - 2u - \frac{1}{u} + C.$ +2.

* 若沒寫 "C" 就
 扣一分.

$= \frac{1}{3} \sin^3 x - 2 \sin x - \frac{1}{\sin x} + C.$ +3.

法 - (2). $\int \frac{x^2}{\sqrt{49-x^2}} dx.$

① $x = 7 \sin \theta.$
 $dx = 7 \cos \theta d\theta.$ +1.

② $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ +1.

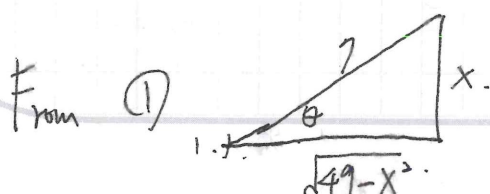
①: $\int \frac{49 \sin^2 \theta}{\sqrt{49-49 \sin^2 \theta}} \cdot 7 \cos \theta d\theta = \int \frac{49 \sin^2 \theta}{7 |\cos \theta|} 7 \cos \theta d\theta.$ +1.

②: $\int \frac{49 \sin^2 \theta}{7 \cos \theta} 7 \cos \theta d\theta = \int 49 \sin^2 \theta d\theta.$ +1.

$= 49 \int \frac{1 - \cos 2\theta}{2} d\theta = \frac{49}{2} (\theta - \frac{1}{2} \sin 2\theta) + C.$ +2.

* 沒寫 "C"
 扣一分.

$= \frac{49}{2} (\theta - \sin \theta \cos \theta) + C = \frac{49}{2} \left(\sin^{-1} \frac{x}{7} - \frac{x \sqrt{49-x^2}}{49} \right) + C.$ +2.



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題號 2, (2)

法二 $\int \frac{x^2}{\sqrt{49-x^2}} dx$ $\begin{cases} x=7\cos\theta \\ dx=-7\sin\theta d\theta \end{cases}$ $0 \leq \theta \leq \pi$

$$= \int \frac{49 \cos^2\theta}{7|\sin\theta|} (-7\sin\theta) d\theta$$

$$\cong -49 \int \cos^2\theta d\theta$$

$$= -49 \int \frac{1+\cos 2\theta}{2} d\theta = \frac{-49}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) + C$$

$$= \frac{-49}{2} \left(\cos^{-1} \frac{x}{7} + \frac{x\sqrt{49-x^2}}{49} \right) + C$$

$$= \frac{-49}{2} \cos^{-1} \frac{x}{7} - \frac{x\sqrt{49-x^2}}{2} + C$$

題號 2(3)

$$\int \frac{2x^3 + 4}{(x-1)^2(x^2+1)} dx$$

$$\frac{2x^3+4}{(x-1)^2(x^2+1)} = \frac{A^0}{(x-1)} + \frac{B^3}{(x-1)^2} + \frac{Cx+D^{2x+1}}{(x^2+1)}$$

$$\Rightarrow 2x^3+4 = A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x-1)^2$$

$$= A(x^3-x^2+x-1) + B(x^2+1) + (Cx+D)(x^2-2x+1)$$

$$= Ax^3 - Ax^2 + Ax - A + Bx^2 + B + Cx^3 - 2Cx^2 + Cx + Dx^2 - 2Dx + D$$

$$\begin{cases} A+C=2 \\ B-A-2C+D=0 \\ A+C-2D=0 \\ B-A+D=4 \end{cases} \Rightarrow \begin{cases} A=0-1 \\ B=3-1 \\ C=2-1 \\ D=1-1 \end{cases}$$

少C扣1

$$\int \frac{3}{(x-1)^2} + \frac{2x+1}{x^2+1} dx = \int \frac{3}{(x-1)^2} + \frac{2x}{x^2+1} + \frac{1}{x^2+1} dx$$

$$= \underline{-3(x-1)^{-1}} + \underline{\ln|x^2+1|} + \underline{\arctan x} + C$$

題號

2(4)

$$\int \frac{\tan x (\sec^2 x + 4 \sec x)}{\sec^2 x + 5 \sec x - 6} dx$$

$$= \int \frac{\tan x \sec x (\sec x + 4)}{\sec^2 x + 5 \sec x - 6} dx$$

$$\int \text{Let } \underline{u = \sec x} \quad \underline{du = \sec x \tan x dx}$$

$$\underline{\frac{u+4}{(u^2+5u-6)}} = \underline{\frac{A}{(u+6)} + \frac{B}{(u-1)}}$$

$$\Rightarrow u+4 = A(u-1) + B(u+6)$$

$$\Rightarrow u=1 \text{ 時 } B = \frac{5}{7} -$$

$$\Rightarrow u=-6 \text{ 時 } A = \frac{2}{7} -$$

$$\int \frac{\frac{2}{7}}{(u+6)} + \frac{\frac{5}{7}}{(u-1)} du$$

$$= \frac{2}{7} \ln|u+6| + \frac{5}{7} \ln|u-1| + C$$

$$= \underline{\frac{2}{7} \ln|\sec x + 6| + \frac{5}{7} \ln|\sec x - 1| + C}$$

少C扣1
未換u扣1
未加絕對值扣1

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題號 2 (5)

2 (5)

$$\int_{-1}^1 \sqrt{x^{10}} + x^2 \tan x \sec x + \frac{1}{1+x^2} dx$$

$$= \int_{-1}^1 \sqrt{x^{10}} dx + \int_{-1}^1 x^2 \tan x \sec x dx + \int_{-1}^1 \frac{1}{1+x^2} dx$$

$$= \int_{-1}^1 |x^5| dx + \int_{-1}^1 x^2 \tan x \sec x dx + \int_{-1}^1 \frac{1}{1+x^2} dx$$

1 pt

$$\int_{-1}^1 |x^5| dx = 2 \cdot \int_0^1 x^5 dx = 2 \cdot \frac{1}{6} x^6 \Big|_0^1 = \frac{1}{3}$$

1 pt

$$\therefore x^2 \text{ is even, } \tan x \text{ is odd, } \sec x \text{ is even}$$

$$\therefore x^2 \tan x \sec x \text{ is odd}$$

1 pt

$$\Rightarrow \int_{-1}^1 x^2 \tan x \sec x dx = 0$$

1 pt

$$\int_{-1}^1 \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_{-1}^1 = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}$$

1 pt

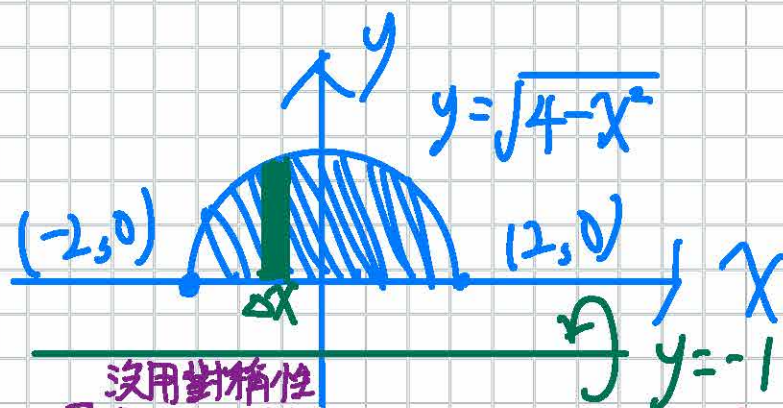
$$\text{Hence } \int_{-1}^1 \sqrt{x^{10}} + x^2 \tan x \sec x + \frac{1}{1+x^2} dx = \frac{1}{3} + 0 + \frac{\pi}{2} = \frac{3\pi+2}{6}$$

1 pt

法1: use Disk (Washer) Method

* 可用對稱方式

$$2 \cdot \int_0^2 \dots$$



$$V = \int_{-2}^2 \pi \left\{ \left[\sqrt{4-x^2} - (-1) \right]^2 - \left[0 - (-1) \right]^2 \right\} dx$$

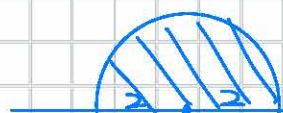
+4分

$$= \int_{-2}^2 \pi \left[4 - x^2 + 2\sqrt{4-x^2} + \cancel{1} - \cancel{1} \right] dx$$

$$= \pi \int_{-2}^2 (4 - x^2) dx + 2\pi \int_{-2}^2 \sqrt{4-x^2} dx$$

(*)

* means 以2為半徑的半圓面積



$$\therefore (*) = \frac{1}{2} \times 2 \times 2 \times \pi = 2\pi$$

+12分

$$\begin{aligned}
&= \pi \left[4x - \frac{1}{3}x^3 \right]_{-2}^2 + 2\pi \times 2\pi \\
&= \pi \times 2 \left(8 - \frac{8}{3} \right) + 4\pi^2 \\
&= 2\pi \times \frac{(24-8)}{3} + 4\pi^2 \\
&= \frac{32}{3}\pi + 4\pi^2
\end{aligned}$$

+1分

+3分

如果算成一個完整 circle 的面積
扣 2 分且不給
最終 answer 的 1 分

評分細節

- ① (A) 列式/積分/化簡錯誤 (b) **通靈** 將不給後面計算分數
② 沒寫 use Disk (Washer) Method **不加分** (除缺少乘入!)

- ③ (*) 的積分可用三角代換操作, 配分如下
沒寫扣一分 (出題 teacher 說必寫)

令 $\left(\begin{array}{l} x = 2\sin\theta \quad \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}], \quad dx = 2\cos\theta \, d\theta \\ \text{When } x=2, \quad 2=2\sin\theta \Rightarrow \sin\theta=1 \Rightarrow \theta=\frac{\pi}{2} \\ x=-2, \quad -2=2\sin\theta \Rightarrow \sin\theta=-1 \Rightarrow \theta=-\frac{\pi}{2} \end{array} \right)$

+1分

$$\text{原式} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{4-4\sin^2\theta} \cdot 2\cos\theta \, d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4\cos^2\theta \, d\theta$$

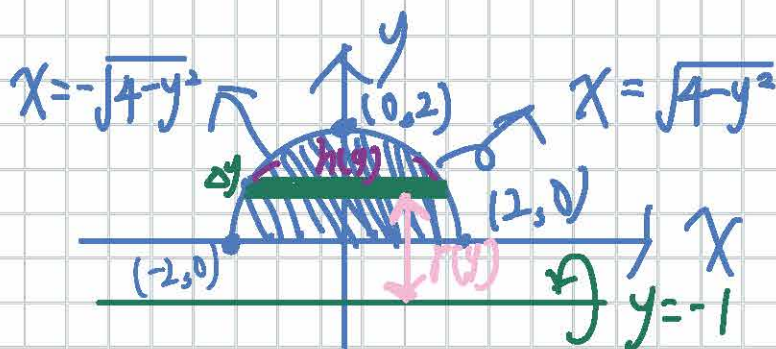
$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2}{1} \cdot \frac{1+\cos(2\theta)}{2} \, d\theta$$

+1分

+1分

$$= 2 \left[\theta + \frac{\sin(2\theta)}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2 \left(2 \times \frac{\pi}{2} + 0 - 0 \right) = \underline{2\pi}$$

題號

3.**法2** Use shell Method

$$\therefore V = \int_0^2 2\pi r(y) h(y) dy$$

$$= \int_0^2 2\pi \cdot (y - (-1)) \cdot (\sqrt{4-y^2} - (-\sqrt{4-y^2})) dy$$

+4分

$$= 4\pi \int_0^2 (y+1) \sqrt{4-y^2} dy$$

$$= 4\pi \left[\underbrace{\int_0^2 y \sqrt{4-y^2} dy}_{\star 1} + \underbrace{\int_0^2 \sqrt{4-y^2} dy}_{\star 2} \right]$$

★1

$$\text{令 } u = 4 - y^2 = u(y)$$

$$du = -2y dy$$

$$u(0) = 4, u(2) = 0$$

★2以2為半徑的 $\frac{1}{4}$ 圓面積

$$\Rightarrow \star 2 = \frac{1}{4} \times 2 \times 2 \times \pi = \pi$$

+1分

$$\begin{aligned}
 &= 4\pi \cdot \int_4^0 u^{\frac{1}{2}} \cdot (-\frac{1}{2}) du + 4\pi \cdot \pi \\
 &= 4\pi \int_0^4 \frac{1}{2} u^{\frac{1}{2}} du + 4\pi^2 \\
 &= 4\pi \cdot \left[\frac{1}{2} \times \frac{2}{3} u^{\frac{3}{2}} \right]_0^4 + 4\pi^2 \\
 &= \frac{32}{3}\pi + 4\pi^2
 \end{aligned}$$

+3分

+1分

+1分

如果算成一個完整或半circle的面積扣2分且不給最終answer的1分

評分細節

- ①^(a) 列式/積分/化簡錯誤 ^(b) **通靈** 將不給後面計算分數
- ② 沒寫 use shell Method **不加分** (除非少乘入!)
- ③ **★2** 的積分可用三角代換操作, 配分如下

沒寫扣一分 (出題teacher說必需)

$$\text{令 } \left(\begin{array}{l} y = 2\sin\theta \\ \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}], \quad dy = 2\cos\theta d\theta \\ \text{When } y = 2, \quad 2 = 2\sin\theta \Rightarrow \sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2} \\ y = -2, \quad -2 = 2\sin\theta \Rightarrow \sin\theta = -1 \Rightarrow \theta = -\frac{\pi}{2} \end{array} \right) +1\text{分}$$

$$\text{原式} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{4-4\sin^2\theta} \cdot 2\cos\theta d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4\cos^2\theta d\theta$$

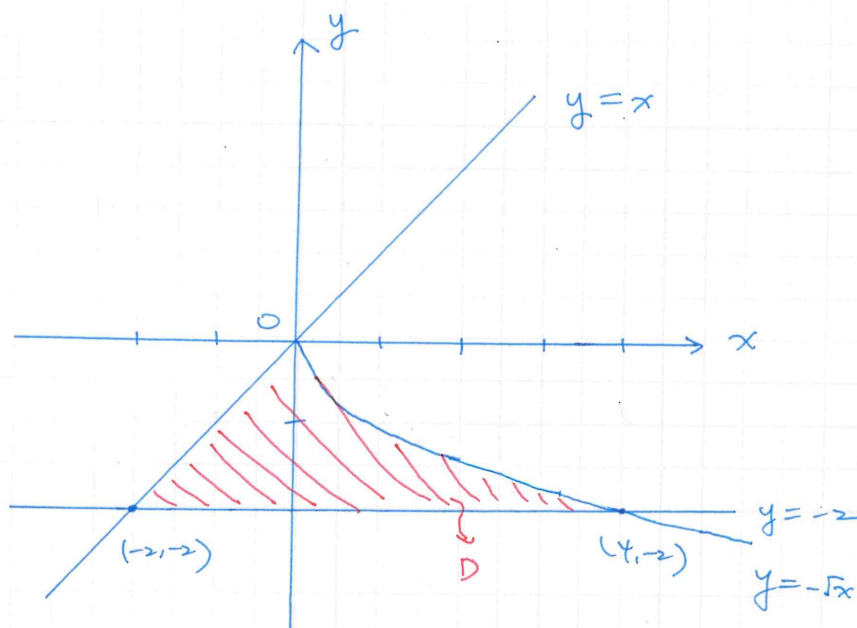
$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2}{2} \cdot \frac{1+\cos(2\theta)}{2} d\theta \quad +1\text{分}$$

$$= 2 \left[\theta + \frac{\sin(2\theta)}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 2(2 \times \frac{\pi}{2} + 0 - 0) = \underline{2\pi} \quad +1\text{分}$$

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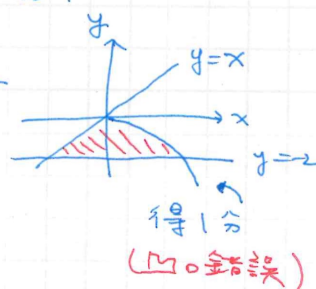
題號 4. (1)

4. (1)
(2pts)

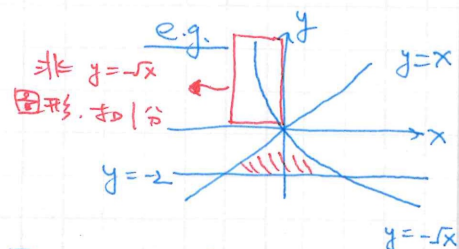


• 此題為繪圖題，未作圖者不予計分。

• $y = -\sqrt{x}$ 之圖形為凹口向上，
畫成凹口向下者得 1 分



• $y = -\sqrt{x}$ 之圖形僅出現在第四象限，
畫出不屬於 $y = -\sqrt{x}$ 圖形部分者扣一分



• D 之區域標示錯誤，但函數圖形正確，得 1 分。

• 其餘部分，圖形大致描繪正確皆可得 2 分。

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題號 4.(2)

$$\textcircled{1} \quad V = \int_{-2}^0 2\pi(-y)(y^2 - y) dy \rightarrow 5 \text{分}$$

$$= -2\pi \left[\frac{1}{4}y^4 - \frac{1}{3}y^3 \right]_{-2}^0 \rightarrow 4 \text{分}$$

$$= \frac{40}{3}\pi \quad \rightarrow 1 \text{分}$$

$$\textcircled{2} \quad V = \int_0^4 \pi [(-2)^2 - (\sqrt{x})^2] dx + \int_{-2}^0 \pi [(-2)^2 - x^2] dx \rightarrow 5 \text{分}$$

$$= \pi \left[4x - \frac{x^2}{2} \right] \Big|_0^4 + \pi \left[4x - \frac{x^3}{3} \right] \Big|_{-2}^0 \rightarrow 4 \text{分}$$

$$= 8\pi + \frac{16}{3}\pi = \frac{40}{3}\pi \rightarrow 1 \text{分}$$

少 2π 扣 3分

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題號

5

$$\frac{dx}{dy} = 2 \cdot \frac{1}{2} (8-y)^{-\frac{1}{2}} (-1) = \frac{-1}{\sqrt{8-y}} \quad 2\text{分}$$

$$S = \int 2\pi x \, ds$$

$$= \int_0^5 2\pi \cdot (2\sqrt{8-y}) \cdot \sqrt{1 + \left(\frac{-1}{\sqrt{8-y}}\right)^2} \, dy \quad 2\text{分}$$

$$= 4\pi \int_0^5 \sqrt{8-y} \sqrt{1 + \frac{1}{8-y}} \, dy$$

$$= 4\pi \int_0^5 \sqrt{9-y} \, dy \quad 2\text{分}$$

let $u = 9-y$, $du = -dy$
 $y=5 \Rightarrow u=4$
 $y=0 \Rightarrow u=9$

$$= 4\pi \int_9^4 \sqrt{u} \, (-du)$$

$$= -4\pi \left(\frac{2}{3} u^{\frac{3}{2}} \right) \Big|_9^4 \quad 2\text{分}$$

$$= \boxed{\frac{152}{3} \pi}$$

2 分

• 有寫對 $x(y) = \frac{-1}{\sqrt{8-y}}$

• 有正確代入公式

• 有正確化簡

積分過程

• 答案正確

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題號 6(a)

$$\int_0^1 \ln x dx = \lim_{a \rightarrow 0^+} \int_a^1 \ln x dx \quad \begin{array}{l} \text{寫成 limit 得 1 分} \\ \text{寫 } a \rightarrow 0 \text{ 不給分} \end{array}$$

$$= \lim_{a \rightarrow 0^+} \left(x \ln x \Big|_a^1 - \int_a^1 1 dx \right) \quad \left. \begin{array}{l} \text{使用分部積分} \\ \text{or 直接寫出 } x \ln x - x \end{array} \right\}$$

$$= \lim_{a \rightarrow 0^+} \left[(x \ln x - x) \Big|_a^1 \right] \quad \text{可得 1 分}$$

$$= \lim_{a \rightarrow 0^+} [a(1 - \ln a) - 1] \quad \text{寫出此式得 1 分}$$

$$\rightarrow = \lim_{a \rightarrow 0^+} [a(1 - \ln a)] - 1 = 0 - 1 = -1 \quad \begin{array}{l} \text{答案對得 1 分} \\ \text{無過程不給分} \end{array}$$

$$\begin{aligned} & \lim_{a \rightarrow 0^+} a(1 - \ln a) \quad (0 \cdot \infty) \\ &= \lim_{a \rightarrow 0^+} \frac{(1 - \ln a)}{a^{-1}} \quad \left(\frac{\infty}{\infty} \right) \\ & \text{L'H} = \lim_{a \rightarrow 0^+} \frac{-\frac{1}{a}}{-\frac{1}{a^2}} = \lim_{a \rightarrow 0^+} a = 0 \end{aligned}$$

算出極限 = 0

得 1 分。

不用 $\lim_{a \rightarrow 0^+}$, 只用 $\int_0^1 \ln x dx = (x \ln x - x) \Big|_0^1 = -1$

只得 2 分

$$\int_0^1 \ln x \, dx$$

$$\text{Let } y = \ln x, \quad dy = \frac{dx}{x}$$

$$\Rightarrow e^y = x \quad \Rightarrow x dy = dx$$

$$\Rightarrow e^y dy = dx$$

$$\begin{cases} x \rightarrow 0^+, & y \rightarrow -\infty \\ x \rightarrow 1^-, & y \rightarrow 0 \end{cases}$$

$$\Rightarrow \int_0^1 \ln x \, dx = \int_{-\infty}^0 y e^y \, dy$$

$$= \lim_{a \rightarrow -\infty} \int_a^0 y e^y \, dy$$

$$= \lim_{a \rightarrow -\infty} (y e^y - e^y) \Big|_a^0$$

$$= \lim_{a \rightarrow -\infty} \left[-1 - [a e^a - e^a] \right]$$

$$= -\lim_{a \rightarrow -\infty} \frac{e^a}{-a} - 1$$

$$= -\lim_{a \rightarrow -\infty} \frac{e^a}{-1} - 1 = -1$$

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題號 6b

<Method 1>

$$\int_0^a \ln x \, dx = \int_1^a \ln x \, dx + \int_0^1 \ln x \, dx = 0$$

$$\int_1^a \ln x \, dx = - \int_0^1 \ln x \, dx$$

$$\int_1^a \ln x \, dx = 1 \rightarrow 2 \text{ points}$$

$$[x \ln x - x] \Big|_1^a = 1$$

$$(a \ln a - 0) - (a - 1) = 1 \rightarrow 1 \text{ point}$$

$$a \ln a - a = 0$$

$$a(\ln a - 1) = 0$$

$$a = e \rightarrow 2 \text{ points.}$$

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題號 6b

<Method 2>

$$\int_0^a \ln x \, dx$$

$$= \lim_{s \rightarrow 0^+} \int_s^a \ln x \, dx$$

$$= \lim_{s \rightarrow 0^+} \left[x \ln x \Big|_s^a - \int_s^a dx \right]$$

$$= \lim_{s \rightarrow 0^+} \left(x \ln x - x \Big|_s^a \right)$$

$$= \lim_{s \rightarrow 0^+} (a \ln a - a) - (s \ln s - s)$$

$$= (a \ln a - a) - \lim_{s \rightarrow 0^+} (\overset{0}{s \ln s} - s) \rightarrow 1 \text{ point}$$

$$= (a \ln a - a) - 0$$

$$a \ln a - a = 0$$

$$a(\ln a - 1) = 0$$

$$a = 0 \text{ or } a = e$$

Since $a > 1$, thus $a = e \rightarrow 2 \text{ points}$

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題號 7 (bonus)

(i) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{7 \cdot 2^7 k^6}{n^7} - \frac{4 \cdot 2^4 k^3}{n^4} + \frac{2}{n} \right) \dots (*)$

$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(7 \cdot 2^7 \cdot \left(\frac{k}{n} \right)^6 - 4 \cdot 2^4 \cdot \left(\frac{k}{n} \right)^3 + 2 \right) \left(\frac{1}{n} \right) \right]$ 1分

$= \int_0^1 \left(7 \cdot 2^7 x^6 - 4 \cdot 2^4 x^3 + 2 \right) dx$ 2分

$= \left. 2^7 x^7 - 2^4 x^4 + 2x \right|_0^1 = 2^7 - 2^4 + 2 = 114$ 2分

(ii)

$(*) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(7 \cdot \left(\frac{2k}{n} \right)^6 - 4 \cdot \left(\frac{2k}{n} \right)^3 + 1 \right) \left(\frac{2}{n} \right) \right]$ 1分

$= \int_0^2 (7x^6 - 4x^3 + 1) dx$ 2分

$= \left. x^7 - x^4 + x \right|_0^2 = 2^7 - 2^4 + 2 = 114$ 2分